

Percentages/Everyday Math

Worksheet 3

1. W16/qp22/q1

- a. The basic price of the 2016 model of a car is \$21000. Sayeed and Rasheed each buy this model of car.
- i. Sayeed pays a deposit of \$756. Calculate the deposit Sayeed pays as a percentage of the basic price. [1]
 - ii. He then pays 24 monthly payments of \$922.25. Calculate the total amount that Sayeed pays as a percentage of the basic price. [2]
- b. Rasheed pays a deposit of \$381 followed by 36 equal monthly payments. The total amount that he pays is 127% of the basic price of \$21000. Calculate Rasheed's monthly payment. [3]
- c. \$21000 represented an increase of 5% on the basic price of the 2015 model. Calculate the difference between the basic prices of the 2015 and 2016 models. [3]

2. S17/qp21/q2

In 2016 Amira's income was \$36720.

- a. This was 2% more than her income in 2015. What was her income in 2015? [2]
- b. In 2016, Amira used her income of \$36720 in the following way. \$12000 was used for rent. $\frac{2}{5}$ of her income was used for food and to pay bills. 15% of her income was spent on leisure. The rest of her income was saved. What percentage of her income did she save? [4]

3. W17/qp21/q1(a)

- i. Jasmine earns \$12.50 for each hour she works. She works for 38 hours each week. She is given a pay increase of 6%. Calculate the total amount Jasmine earns each week after the pay increase. [3]
- ii. Abdul earns \$525 each week. He moves to a new job where he earns \$462 each week. Calculate the percentage reduction in his earnings in his new job. [2]
- iii. Maria is given a pay increase of 3%. After the pay increase, she earns \$2472 each month. Calculate her monthly pay before the pay increase. [2]

4. W17/qp22/q1

- a. Sara buys a new car. The cash price of the car is \$4500. She can pay for the car using option A or option B.

Option A
Pay $\frac{1}{5}$ of the cash price then 12 monthly payments of \$340

Option B
Pay 12% of the cash price then 24 monthly payments of \$195

Which option is cheaper and by how much?

[4]

- b. Sara's car uses 5.2 liters of petrol for each 100km she drives. Petrol costs \$0.85 per liter. Sara drives 240 km. Calculate the cost of the petrol used for this journey. Give your answer correct to the nearest cent. [3]
- c. Sara pays a total of \$322 for her car insurance. This total is made up of a basic charge plus 15% sales tax. Calculate the amount of sales tax that Sara pays. [3]

5. W18/qp21/q2(c,d)

- a. Dina sells a shirt for \$15.66. The price includes 8% sales tax. Calculate the cost of the shirt excluding sales tax. [2]
- b. Dina is saving to make improvements to her shop. She invests \$3500 for 4 years at 1.7% per year compound interest. Calculate the value of Dina's investment after 4 years. Give your answer correct to the nearest cent. [3]

6. W19/qp22/q1

Tanya owns a small business.

- a. Tanya has 4 employees. Every week, the employees each work for $7\frac{3}{4}$ hours each day for 5 days. Each employee is paid \$15.20 per hour. Calculate the total amount Tanya pays her 4 employees in one week. [2]
- b. The business made a profit of \$25700 in 2017 compared with \$22102 in 2018. Calculate the percentage decrease in profit from 2017 to 2018. [2]
- c. Tanya must add 8% sales tax to the initial cost of a job. She then adds 15% to the cost, including sales tax, to find the amount to charge a client. Tanya charges one client a total of \$465.75 for a job. Calculate the initial cost of this job. [3]
- d. Tanya invests \$8500 in an account paying 3.1% per year compound interest. At the end of 5 years, she takes \$9300 from the account to buy new equipment for the business. Calculate how much money is left in the account after buying the new equipment. [3]

7. S20/qp21/q4

Anton invests \$6000 in an account for 5 years. The account has a compound interest rate of 2.5% per year. At the end of 5 years, he spends \$4200 of this money on a family holiday to Malaysia.

- a. How much money is left in the account? [3]
- b. Anton changes \$800 into Malaysian Ringgits (MYR) for his trip. The exchange rate is \$1 = 3.16 MYR. He spends 2250 MYR and then changes the remaining money back into dollars (\$). The exchange rate on his return is \$1 = 3.27 MYR. How many dollars does he receive on his return? Give your answer correct to the nearest dollar. [3]
- c. Anton invests \$1500 in another account. The account has a compound interest rate of $p\%$ per year. At the end of 3 years, there is \$1598.85 in the account. Calculate p . Give your answer correct to 2 decimal places. [3]

8. W20/qp21/q1

Here is some information about a holiday.

7-night holiday \$340 per person
8% discount if you book before 31 March

- a. On 15 February, Naseem books this holiday for 2 people.
Calculate the total cost of his holiday. [2]
- b. Naseem hires a car for his holiday. The total cost is \$241.50. This cost includes 15% tax.
Calculate the cost of hiring the car excluding tax. [2]
- c. Naseem drives a total of 800 km on holiday. He uses a total of 29.6 liters of fuel. Calculate the average rate of fuel used in liters per 100km. [2]
- d. Naseem changes \$450 to euros (€) for his holiday. The exchange rate between dollars and euros is \$1 = €0.82. On holiday, he spends €297. Naseem changes the remaining money back to dollars when he returns home. The exchange rate is now \$1 = €0.80. Work out how many dollars he receives. [3]

9. S21/qp22/q1

- a. The price of an electric drill is \$78. In a sale, the price is reduced by 15%. Calculate the sale price. [2]
- b. The exchange rate between dollars (\$) and euros (€) is \$1 = €0.85. Michael changes \$100 to euros. He buys a clock costing €58.99. He changes the remaining money back to dollars. Calculate the amount, in dollars, he has left. [2]

ACE SIMPLE

Simple interest at
2.1% per year

COOL COMPOUND

Compound interest at
2% per year

- c. Pietro invests \$3500 in the Ace Simple account for 4 years. Eliana invests \$3500 in the Cool Compound account for 4 years. At the end of the 4 years, who has more money in their account and by how much? [4]

Answer Key – Worksheet 3

1. (a)(i)3.6 (ii)109 (b)730.25 (c)1000
2. (a)36000 (b)12.3
3. (a)(i)503.5 (ii)12 (iii)2400
4. (a) A by 240 (b)10.61 (c)42
5. (a)14.5 (b)3744.14
6. (a)2356 (b)14 (c)375 (d)601.75 or 601.76
7. (a)2590 or 2588 (b)85 (c)21.5
8. (a)625.6 (b)210 (c)3.7 (d)90
9. (a)66.3 (b)30.6 (c)Pietro AND 5.48 or 5.49



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